INCH-POUND

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DETAIL SPECIFICATION

MOUNT, TELESCOPE AND QUADRANT - M187A1

This specification is approved for use by ARDEC and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Mount, Telescope and Quadrant: M187A1 which supports the Telescope, Panoramic: M137A2 for use on the M119 series of towed howitzers. The mount provides elevation adjustment and compensates for angles of cant when laying the guns for indirect fire.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 and 4 of this specification, whether or not documents are listed in this section.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-F-13926	-	Fire Control Materiel, Manufacture, and Inspection, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of federal and military specifications, standards, and handbooks are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

8213745	-	Adapter, Fixture for Mt. Telescope M137, Quad. Elevation M15
8565556	-	Pressure Tester – Two Station
8658940	-	Inspection Aid
10555620	-	Sighting Device
11747889	-	Gage, Interchangeability Max
12984689	-	Mount, Telescope and Quadrant: M187A1
12984713	-	Telescope, Panoramic: M137A2
13020787	-	Plate, Identification
13028309	-	Adapter, Fixture

(Copies of these drawings may be requested online at usarmy.picatinny.ardec.list.drawing-request-help-desk@mail.mil or from U.S. Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000)

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2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.3 Mount. The mount shall conform to drawing 12984689.

3.4 Environmental.

3.4.1 Storage temperature. While thermally stabilized at temperatures of +160 ± 5°F and -60 ± 5°F, the mount shall not exhibit any signs of physical failure or damage. Upon return to and stabilization at standard ambient temperature, +60°F to +90°F, the mount shall meet the requirements of 3.5 through 3.11 inclusive.

3.4.2 Operating temperature. The mount shall meet the requirements of 3.7 while exposed to and thermally stabilized at temperatures of +145 ± 5°F and -50 ± 5°F. Upon return to and stabilization at standard ambient temperature, +60°F to +90°F, the mount shall meet the requirements of 3.5 through 3.11 inclusive.

3.4.3 Shock. With a M137A2 panoramic telescope, 12984713, attached in the appropriate location, the mount shall withstand a total of 18 shock impulses along the 3 mutually perpendicular axes as defined in Figure 1. Each shock impulse shall be a half sine wave with a time duration of 0.010 ± 0.001 seconds and peak amplitude for each shock impulse as shown in Table I when measured at the face of the counterbox of the M137A2 telescope (see Figure 1). Subsequent to shock, the mount shall show no evidence of damage or physical failure and shall meet the requirements of 3.5 through 3.11 inclusive.

TABLE I. Shock orientation and impulse at M137A2 counterbox.

Orientation	Shock impulse and direction	Number of impulses
Longitudinal	-100 g's, +100 g's	3 each direction
Vertical	-100 g's, +100 g's	3 each direction
Transverse	-60 g's, +60 g's (See 6.7)	3 each direction

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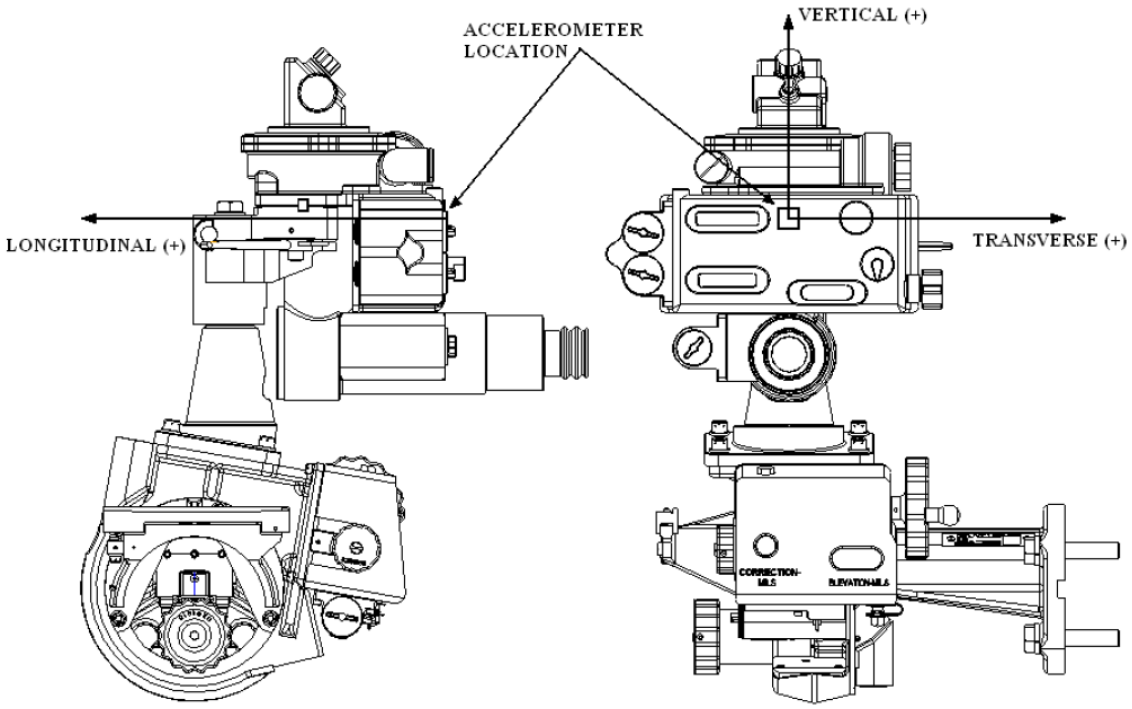


FIGURE 1. Orientation of shock loads on M187A1 mount.

3.4.4 Vibration "A". The mount shall withstand a total of 270 ± 5 minutes of sweep-cycle vibration. The vibration shall be applied for 90 ± 2 minutes along each of the three mutually perpendicular major axes shown in Figure 1. A complete sweep-cycle shall consist of vibration from origin (5 Hz at 1 inch double amplitude) to mid-point (5 ± 0.5 g's at 500 Hz) to origin, and shall have a duration of 15 ± 1 minutes. Double amplitude shall be constant at 1 inch between 5 Hz and 10 Hz, and varied with frequency to maintain a constant 5 ± 0.5 g's acceleration between 10 Hz and 500 Hz. Upon completion of vibration, the mount shall exhibit no evidence of damage or physical failure and shall meet the requirements of 3.5 through 3.11 inclusive.

3.4.5 Vibration "B". The mount shall be vibrated in the vertical plane shown in Figure 1 for a total of 15 ± 1 minutes of sweep cycle vibration. A complete sweep-cycle shall consist of vibration from origin (5 Hz at 1 inch double amplitude) to mid-point (5 ± 0.5 g's at 500 Hz) to origin, and shall have a duration of 15 ± 1 minutes. Double amplitude shall be constant at 1 inch between 5 Hz and 10 Hz and varied with frequency to maintain a constant 5 ± 0.5 g's acceleration between 10 Hz and 500 Hz. Upon completion of vibration, the mount shall exhibit no evidence of damage or physical failure and shall meet the requirements of 3.5 through 3.11 inclusive.

3.5 Sealing. The counter box enclosure of the mount shall be capable of holding dry nitrogen at a minimum internal pressure of 2.0 psi for a minimum of 2 hours without the loss of pressure in excess of 0.2 psi for that 2 hour period.

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3.6 Purging. The counter box enclosure of the mount shall be purged with dry nitrogen supplied at a pressure of 5.0 ± 0.2 psig until the dew point of the emergent gas is no higher than -25°F. Purging apparatus shall be removed when the relief valve is reseated and maintains a pressure between 0.25 and 0.5 psig.

3.7 Performance.

3.7.1 Elevation travel. From a level starting position, the mount shall be able to travel through a minimum excursion of 1333 mils in elevation and a minimum excursion of 89 mils in depression.

3.7.2 Elevation plumb travel. From a level starting position, the line of sight shall not deviate from a plumb line by more than 0.5 mil on either side and the cross level vial bubble shall remain level within the thickness of a graduation line, when the elevation knob is actuated through its full range of movement.

3.7.3 Elevation error. Elevation error shall not exceed 0.5 mil, excluding backlash, at any elevation between -89 and +1333 mils.

3.7.4 Elevation backlash. Backlash in the elevation mechanism shall not exceed 0.5 mil.

3.7.5 Cross level (cant) travel. From a level starting position, the mount shall be able to cant a minimum of 356 mils to the right and a minimum of 356 mils to the left using the cross level knob.

3.7.6 Cross level (cant) backlash. Backlash in the cross level mechanism shall not exceed 0.5 mil.

3.7.7 Correction counter operation. The correction counter shall travel through an excursion of +9 mils to -9 mils. Plus (+) values set on the correction counter shall subtract an equal number of mils from the elevation counter. Minus (-) values set on the correction counter shall add an equal number of mils to the elevation counter.

3.7.8 Correction counter accuracy. The angle of the level bar shall correspond to the value displayed on the correction counter within 0.2 mil throughout the full range of the correction counter (-9 mils through +9 mils, inclusive) when the mount is reoriented to the initial elevation reading.

3.7.9 Counter alignment. Variation in the alignment of the number centerline in the elevation counter shall be within one sixteenth inch (1/16”).

3.7.10 Cant correction. The mount shall make left and right cant corrections at various elevations in accordance with the values in Table II. The elevation counter shall read within plus or minus 0.5 mils of the elevation angle of the level bar.

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TABLE II. Cant correction.

Counter elevation (mils)	Cant angle (mils)	Correction (mils)	Correction tolerance (mils)
500	177.8 ± 0.5	96.1	0.5
800	177.8 ± 0.5	180.5	1.0
1000	177.8 ± 0.5	271.9	2.0

3.7.11 Knob torque. The running torque required to operate the knobs shall conform to the values in Table III at the corresponding temperatures.

TABLE III. Knob torque.

Knob	Torque between +60°F to +90°F and at +145 ± 5°F	Torque at °F -50 ± 5°F
Cross level	4 to 12 inch pounds	25 inch pounds maximum
Elevation level	4 to 14 inch pounds	25 inch pounds maximum
Correction	2 to 4 inch pounds	8 inch pounds maximum

3.8 Mount rigidity. With the mount secured by the mounting points and with a telescope attached, the total lateral movement when a 20 pound load is applied and released from one side of the telescope mounting flange and then applied and released from the opposing side shall not exceed the values of Table IV at each of the specified elevations.

TABLE IV. Mount rigidity.

Elevation (mils)	Cant angle (mils)	Total lateral movement (mils)
0	0	0.25
800	0	0.75
1100	0	1.75
1333	0	3.50

3.9 Illumination. When illuminated using the mount's power and lighting system, the level vial bubbles, graduation lines, and two counters shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into darkness with and without the use of a 50% neutral density filter.

3.10 Interchangeability. The mounting surface on the M187A1 mount for the M137A2 telescope shall be able to accept a M137A2 telescope with its mounting surfaces at maximum material condition.

3.11 Workmanship. Workmanship of the M187A1 mount shall be in accordance with the requirements of MIL-F-13926.

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4. VERIFICATION

TABLE V. Requirement/verification cross reference matrix.

Method of Verification 1 - Analysis 2 - Demonstration 3 - Examination 4 – Test						Class of Verification A - First Article B - Conformance		
Section 3 Requirements	Description	Verification Method				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1	First article			X	X	X		4.2
3.2	Conformance			X	X		X	4.3
3.3	Mount			X	X	X	X	4.3.2.4
3.4.1	Storage temperature				X	X	X	4.4.1
3.4.2	Operating temperature				X	X	X	4.4.2
3.4.3	Shock				X	X	X	4.4.3
3.4.4	Vibration “A”				X	X		4.4.4
3.4.5	Vibration “B”				X		X	4.4.5
3.5	Sealing				X	X	X	4.5
3.6	Purging				X	X	X	4.6
3.7.1	Elevation travel				X	X	X	4.7.1
3.7.2	Elevation plumb travel				X	X	X	4.7.2
3.7.3	Elevation error				X	X	X	4.7.3
3.7.4	Elevation backlash				X	X	X	4.7.4
3.7.5	Cross level (cant) travel				X	X	X	4.7.5
3.7.6	Cross level (cant) backlash				X	X	X	4.7.6
3.7.7	Correction counter operation				X	X	X	4.7.7
3.7.8	Correction counter accuracy				X	X	X	4.7.8
3.7.9	Counter alignment				X	X	X	4.7.9
3.7.10	Cant correction				X	X	X	4.7.10
3.7.11	Knob torque				X	X	X	4.7.11
3.8	Mount rigidity				X	X	X	4.8
3.9	Illumination				X	X	X	4.9
3.10	Interchangeability				X	X	X	4.10
3.11	Workmanship			X		X	X	4.11

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article. When specified, a sample shall be subjected to first article verification in accordance with Table VI.

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TABLE VI. First article inspection.

Examination or Test 1/	Section 3 Requirement	Section 4 Verification
Mount	3.3	Visual
Storage temperature	3.4.1	4.4.1
Operating temperature	3.4.2	4.4.2
Shock	3.4.3	4.4.3
Vibration “A”	3.4.4	4.4.4
Sealing	3.5	4.5
Purging	3.6	4.6
Elevation travel	3.7.1	4.7.1
Elevation plumb travel	3.7.2	4.7.2
Elevation error	3.7.3	4.7.3
Elevation backlash	3.7.4	4.7.4
Cross level (cant) travel	3.7.5	4.7.5
Cross level (cant) backlash	3.7.6	4.7.6
Correction counter operation	3.7.7	4.7.7
Correction counter accuracy	3.7.8	4.7.8
Counter alignment	3.7.9	4.7.9
Cant correction	3.7.10	4.7.10
Knob torque	3.7.11	4.7.11
Mount rigidity	3.8	4.8
Illumination	3.9	4.9
Interchangeability	3.10	4.10
Workmanship	3.11	4.11
1/ With the exception of the verifications that are required to be completed at the operating temperatures during the verification of the operating temperature requirement, the verification of requirements 3.5 through 3.11 inclusive do not have to be confirmed after each of the required environmental tests; if preferred by the personnel conducting the testing and in an effort to reduce verification time, the verification of requirements 3.5 through 3.11 inclusive may be conducted after all of the required environmental tests have been completed on the unit under test. At a minimum, the verification of requirements 3.5 through 3.11 inclusive must be conducted after all of the environmental testing has been completed on the unit under test. Otherwise, the order in which the requirements are verified may be selected at the discretion of the personnel conducting the testing.		

4.2.1 First article quantity. First article inspections shall be performed on three (3) complete units.

4.2.2 First article inspection to be performed. The first article inspection shall be performed in accordance with Table VI.

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4.2.3 First article rejection. If any item of the sample fails to comply with any of the first article requirements, the first article shall be rejected.

4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with the lot formation requirements of MIL-STD-1916, paragraph "Formation and identification of lots and batches."

4.3.2 Classification of characteristics.

4.3.2.1 Conformance inspection quantity. Conformance inspection quantities shall be in accordance with 4.3.2.4 and MIL-STD-1916, paragraph "Sampling of lots or batches." Conformance criteria shall be in accordance with MIL-STD-1916, paragraph "Verification level (VL)."

4.3.2.2 Conformance inspection to be performed. The conformance inspection shall be performed in accordance with 4.3.2.4.

4.3.2.3 Conformance rejection. If any item fails to comply with the applicable conformance inspection requirements, the lot shall be rejected.

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4.3.2.4	Mount.			Drawing number 12984689
Classification	Examination or test 1/	Conformance criteria	Requirement paragraph	Next higher assembly None Inspection method reference
Major				
101	Presence of Plate, Identification, 13020787	100%	3.3	Visual
102	Storage temperature	VL-IV	3.4.1	4.4.1
103	Operating temperature	100%	3.4.2	4.4.2
104	Shock	100%	3.4.3	4.4.3
105	Vibration “B”	100%	3.4.5	4.4.5
106	Sealing	100%	3.5	4.5
107	Purging	100%	3.6	4.6
108	Elevation travel	100%	3.7.1	4.7.1
109	Elevation plumb travel	100%	3.7.2	4.7.2
110	Elevation error	100%	3.7.3	4.7.3
111	Elevation backlash	100%	3.7.4	4.7.4
112	Cross level (cant) travel	100%	3.7.5	4.7.5
113	Cross level (cant) backlash	100%	3.7.6	4.7.6
114	Correction counter operation	100%	3.7.7	4.7.7
115	Correction counter accuracy	100%	3.7.8	4.7.8
116	Counter alignment	100%	3.7.9	4.7.9
117	Cant correction	100%	3.7.10	4.7.10
118	Knob torque	100%	3.7.11	4.7.11
119	Mount rigidity	100%	3.8	4.8
120	Illumination	100%	3.9	4.9
121	Interchangeability	100%	3.10	4.10
Minor				
201	Workmanship	VL-II	3.11	4.11

1/ With the exception of the verifications that are required to be completed at the operating temperatures during the verification of the operating temperature requirement, the verification of requirements 3.5 through 3.11 inclusive do not have to be confirmed after each of the required environmental tests; if preferred by the personnel conducting the testing and in an effort to reduce verification time, the verification of requirements 3.5 through 3.11 inclusive may be conducted after all of the required environmental tests have been completed on the unit under test. At a minimum, the verification of requirements 3.5 through 3.11 inclusive must be conducted after all of the environmental testing has been completed on the unit under test. Otherwise, the order in which the requirements are verified may be selected at the discretion of the personnel conducting the testing.

4.4 Environmental. The temperature tests of 4.4.1 and 4.4.2 can be combined into one test procedure and performed as such. It is the contractor’s responsibility to assure that all required inspections are performed at the specified temperature ranges.

With the exception of the verifications that are required to be completed at the operating temperatures during 4.4.2, the verification of requirements 3.5 through 3.11 inclusive do not have

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to be confirmed after each of the required environmental tests; if preferred by the personnel conducting the testing, the verification of requirements 3.5 through 3.11 inclusive may be conducted after all of the required environmental tests have been completed on the unit under test. At a minimum, the verification of requirements 3.5 through 3.11 inclusive must be conducted after all of the environmental testing has been completed on the unit under test.

4.4.1 Storage temperature. The mount, with fresh batteries installed, shall be placed in the test chamber and the temperature of the chamber reduced gradually to -60 ± 5°F and shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period, the mount shall be removed from the environmental chamber and inspected for signs of physical failure or damage. The mount shall be placed back in the environmental chamber and the temperature shall then be gradually raised to +160 ± 5°F and the mount shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period, the mount shall be removed from the environmental chamber and inspected for signs of physical failure or damage. The mount shall be placed back in the environmental chamber and the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60°F to +90°F) at which time the mount shall be removed from the test chamber. The mount shall then meet the requirements of 3.5 through 3.11 inclusive. (See 6.6)

4.4.2 Operating temperature. The mount, with fresh batteries installed, shall be placed in the environmental chamber and the temperature of the chamber reduced gradually to -50 ± 5°F. The mount shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period, and while at -50 ± 5°F, the mount shall meet the requirements of 3.7. Upon completion of this portion of the test, the mount shall then be placed back in the environmental chamber and the temperature shall be raised gradually to +145 ± 5°F. The mount shall remain at this temperature for a period of 4 hours. At the completion of this 4 hour period, and while at +145 ± 5°F the mount shall meet the requirements of 3.7. Upon completion of this test, the mount shall be returned to the environmental chamber and the temperature shall be gradually reduced to room ambient temperature (+60°F to +90°F) at which time the mount shall be removed from the test chamber and shall meet the requirements of 3.5 through 3.11 inclusive. (See 6.6)

4.4.3 Shock. The M187A1 mount shall be mounted to the test equipment utilizing the keyway and four mounting bolts torqued to 85-90 foot-pounds, and a M137A2 telescope, 12984713, or equivalent weight and center of gravity simulators shall be mounted to it in the designated location, and then it shall be subjected to the shock requirements in accordance with 3.4.3 with one exception; for conformance testing only, the transverse shock loads used for verification shall be -20 g's and +20 g's instead of the -60 g's and +60 g's required. (See 6.7) Upon completion of the shock test, the mount shall meet the requirements of 3.5 through 3.11 inclusive.

4.4.4 Vibration "A". The mount shall be mounted utilizing the keyway and four mounting bolts torqued to 85-90 foot-pounds and then subjected to the vibration requirements in accordance with 3.4.4. Upon completion of the vibration test, the mount shall meet the requirements of 3.5 through 3.11 inclusive.

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4.4.5 Vibration "B". The mount shall be mounted utilizing the keyway and four mounting bolts torqued to 85-90 foot-pounds and then subjected to the vibration requirements in accordance with 3.4.5. Upon completion of the vibration test, the mount shall meet the requirements of 3.5 through 3.11 inclusive.

4.5 Sealing. Accuracy of the test equipment shall be equal to or exceed the accuracies depicted on Pressure Tester – Two Station Drawing 8565556. Prior to the start of the test, secure the poppet valve in a closed position. Pressurize the counter box with dry nitrogen to a minimum of 2 psig and then remove the pressure source. Observe interior pressure of counter box. After a period of at least 2 hours, observe the interior pressure of the counter box and compare with starting pressure. Any observed change in pressure shall be within the limits stated in 3.5.

4.6 Purging. Accuracy of the test equipment shall be equal to or exceed the accuracies depicted on Pressure Tester – Two Station Drawing 8565556. Connect the test equipment to the purging valve of the counter box. Adjust the dry nitrogen pressure to 6.00 ± 0.25 psig. The pressure is to remain constant until the dew point of the emergent gas is no higher than -25°F. Purging apparatus shall then be removed and the relief valve shall reseat and maintain an internal pressure within the limits stated in 3.6.

4.7 Performance. A cross level fixture equipped with fixture adapter 8213745 or 13028309 shall be used for the inspection of M187A1 mount. Position the test fixture on a vibration free surface in accordance with the set-up instructions outlined on drawing 8213745. Follow all set-up and operating instructions prior to securing the mount to the fixture. Surfaces referenced are defined in Figure 2. Verifications shall be done with the starting position of the mount as follows:

a. Surface "A" shall be in a vertical plane within 0.05 mils.

b. Keyway "B" shall be in a horizontal plane within 0.05 mils.

c. Surface "C" shall be in a horizontal plane within 0.5 mils.

d. Surface "C" shall be in parallel plane with Keyway "B" within 0.5 mils.

e. The four mounting bolts shall be torqued to 85 to 90 foot pounds.

f. The Elevation Counter shall read 0000 with backlash removed. The Correction Counter shall be set to 0.

g. The elevation and cross level vial bubbles shall be centered within the thickness of a graduation line.

h. Surface "D" shall be parallel to Keyway "B" within 0.2 mil.

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i. The locating keys shall be perpendicular to Surface “A” within 0.4 mil.

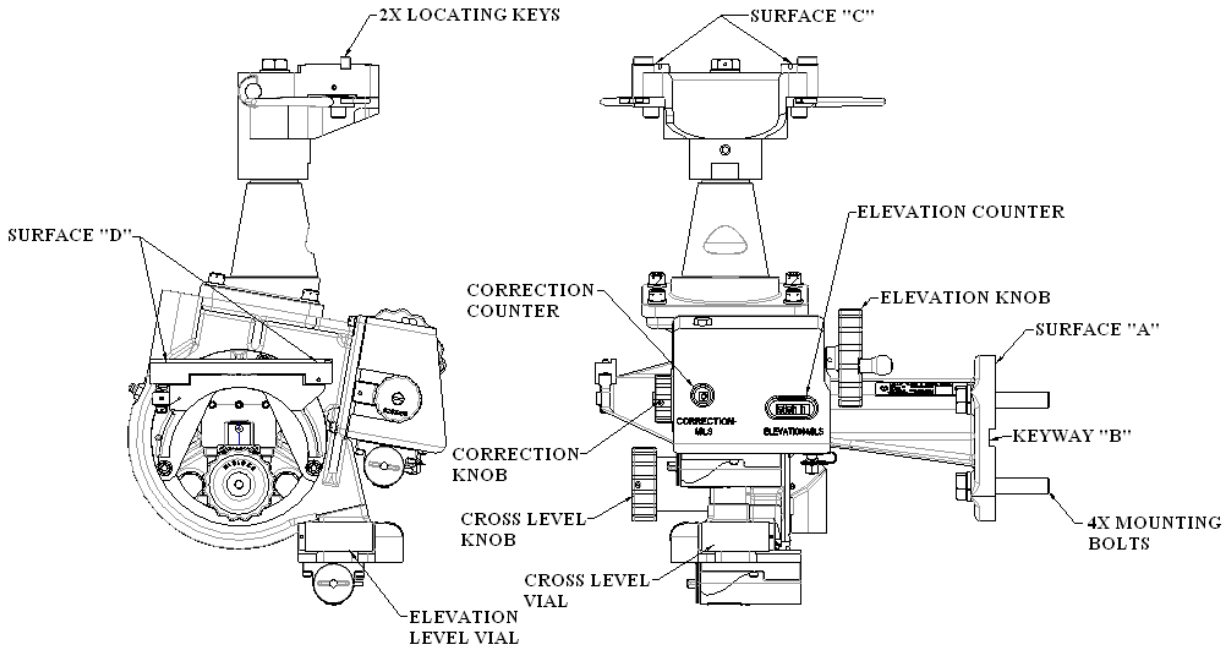


FIGURE 2. Components and surfaces defined on M187A1 mount.

4.7.1 Elevation travel. With the mount and fixture positioned as outlined in 4.7, rotate the mount’s elevation knob to reflect 1333 mils on the elevation counter. Elevate the fixture to center the elevation level vial bubble and confirm the elevation with Inspection Aid (Gunner’s Quadrant), 8658940, on surface “D” of Figure 2 (the level bar). Repeat this procedure for -89 mils depression. The limits of travel shall be within the limits of 3.7.1

4.7.2 Elevation plumb travel. With the mount and fixture positioned as outlined in 4.7, and with an appropriate sighting device assembled to the mount, bring the intersecting point of plumbed target centerlines into coincidence with the sighting device’s optical axis. Depress the mount and fixture to -89 mils, without adjusting cant. The cant level vial bubble and line of sight shall be within the limits of 3.7.2. Bring the mount and fixture back to the orientation of 4.7. Sight through an appropriate sighting device and bring the intersecting point of plumbed target centerlines into coincidence with the sighting device’s optical axis. Elevate the mount and fixture to 1333 mils, without adjusting cant. The cant level vial bubble and line of sight shall be within the limits of 3.7.2.

4.7.3 Elevation error. With the fixture and mount positioned as outlined in 4.7, and with an appropriate sighting device assembled to the mount, insert a whole number mount elevation into the mount, removing backlash. Rotate the fixture until the mount elevation level vial bubble is centered. Using Inspection Aid (Gunner’s Quadrant) 8658940 on surface “D” of Figure 2 (the level bar), measure the actual elevation. Repeat at other elevations for a total of 6 different elevation readings. The difference between the whole number mount elevation and the actual reading from the Gunner’s Quadrant shall be within the limits outlined in 3.7.3.

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4.7.4 Elevation backlash. With the mount and fixture positioned as outlined in 4.7, and with Sighting Device, 10555620, attached, place the Inspection Aid (Gunner's Quadrant), 8658940, on the appropriate sets of pads on the sighting device and take an elevation reading with the Gunner's Quadrant. Elevate the mount by rotating the mount elevation knob approximately one revolution clockwise. Rotate the knob in counterclockwise back to the original whole number elevation, making sure not to override this position. Place the Gunner's Quadrant on the pads of the sighting device and take an elevation reading with the Gunner's Quadrant. The difference between the first Gunner's Quadrant reading and the second is considered backlash and shall not exceed the limit outlined in 3.7.4.

4.7.5 Cross level (cant) travel. With the mount positioned on the fixture as outlined in 4.7, rotate the cant mechanism to the full extent of travel. Place the Inspection Aid (Gunner's Quadrant), 8658940, on surface "C" of Figure 2 and read the cant angle. This reading must be equal to or exceed the minimum requirement outlined in 3.7.5. Repeat this procedure at the opposite extreme setting.

4.7.6 Cross level (cant) backlash. With the mount positioned on the fixture as outlined in 4.7 making sure that the backlash has been removed from the cross leveling knob using a clockwise motion to come to the position without overtravel, mark a line with tape on the cross leveling knob and the adjacent housing and obtain a cross level angle reading using the Inspection Aid (Gunner's Quadrant), 8658940, on surface "C" of Figure 2. Record this reading. Continue to rotate the cross leveling knob one revolution clockwise beyond the index mark and return back to the original index mark counterclockwise without over traveling the mark. Record this reading using the inspection aid. The difference between these two inspection aid readings is backlash and shall be within the limits of 3.7.6.

4.7.7 Correction counter operation. With the mount positioned on the fixture as outlined in 4.7, rotate the correction counter knob to indicate that the maximum correction range is within the limits of 3.7.7 in both directions. Rotate correction counter in the positive direction through the 9 mil range, while observing the elevation counter to verify that it has decreased the same number of mils as the correction counter has increased. Repeat the above procedure while rotating the correction counter in the negative direction through the same range and verifying the proper increase on the elevation counter.

4.7.8 Correction counter accuracy. With the mount positioned on the fixture as outlined in 4.7, rotate the correction counter to the minimum setting of -9 mils as specified in 3.7.8. Return the elevation counter to zero with the elevation knob of the mount. By means of the fixture, level the elevation level vial of the mount. Do not override this level position. Place the Inspection Aid (Gunner's Quadrant), 8658940, on surface "D" of Figure 2 (level bar) and obtain elevation reading. This reading shall be identical to the correction counter reading within the limits of 3.7.8. Repeat this procedure at the setting of +9 mils as specified in 3.7.8.

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4.7.9 Counter alignment. Actuate the elevation knob through its complete travel range. During this travel make a minimum of 2 stops on each side of zero elevation, and by standard measuring equipment, determine the variation between the numeral centerlines. Digit alignment shall be within the limits outlined in 3.7.9.

4.7.10 Cant correction. With the mount and fixture positioned as outlined in 4.7, and with the auxiliary telescope's sighting device assembled to the mount, rotate the auxiliary telescope's line of sight to 1600 mils and engage the locating pin in the proper locating hole indicated in the set up instructions of 11730831. The line of sight shall coincide with the vertical target line. Elevation angle of the level bar of the mount shall coincide with the elevation reading shown on the elevation counter within the limits of 3.7.10. Elevate and cant the fixture to selected set of values in Table II and depress and cant the mount an equal amount in the opposite direction to center the cant and elevation level vial bubbles. The telescope's line of sight shall coincide with the target vertical centerline within the limits outlined in 3.7.10.

4.7.11 Knob torque. All torque readings specified in 3.7.11 shall be measured using standard torque wrench and appropriate adapters. All readings shall conform to the values specified in Table III of 3.7.10 for the temperature specified.

4.8 Mount rigidity. With the mount positioned as specified in 4.7 with an appropriate sighting device affixed to the top mounting surface and zeroed on a plumbed target. Gradually apply and release a 20 pound load at the top left side of the telescope support in line with the keyway as shown in Figure 3. Take the reading on the test target and record the position. Repeat this procedure gradually applying and releasing the specified load at the top right side of the telescope support in line with the keyway. Take the reading on the test target and record the movement from the previously recorded position. If the two displacement values are on opposite sides of the original azimuth line, the total lateral movement is the summation of the absolute values of the displacement; if both values are on the same side of the original azimuth line, the lateral movement is the greater of the two. Repeat at 800 mils, 1100 mils, and 1333 mils elevation to conform to the requirements of 3.8 and Table IV. The total lateral movement at each of the elevations shall not exceed the values specified in Table IV of 3.8.

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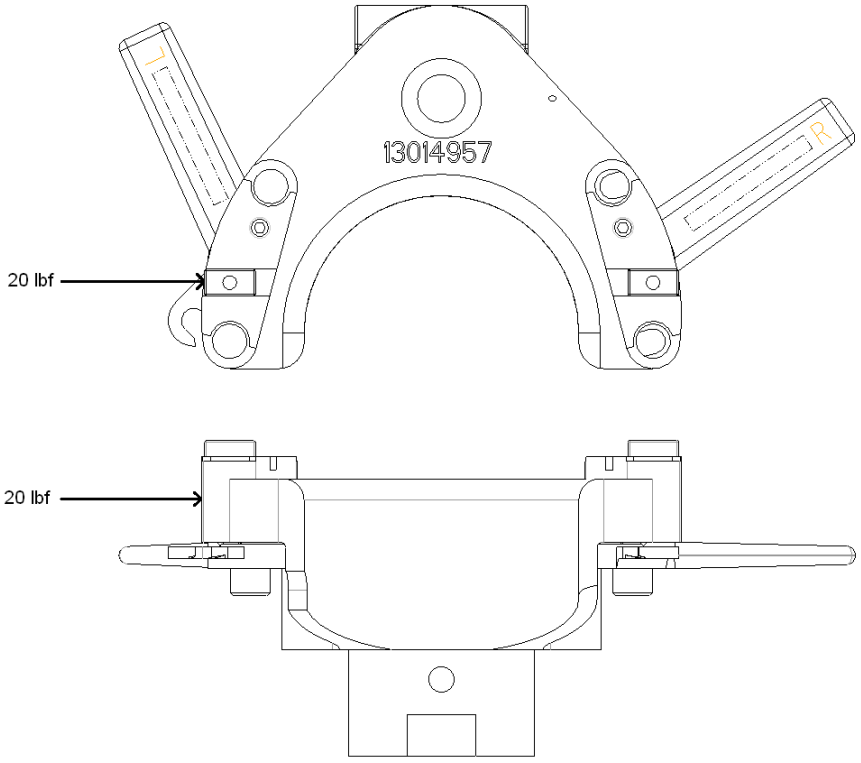


FIGURE 3. Proper load location for mount rigidity test on M187A1 mount.

4.9 Illumination. Place a 50% neutral density filter in front of the counters and level vials. In a darkened room, visually inspect for conformance with the requirements of 3.9. Remove the filter and repeat the visual inspection to verify conformance with 3.9.

4.10 Interchangeability. Interchangeability shall be verified by the insertion, seating, securing, and subsequent removal of maximum interchangeability gage, 11747889, on the mounting surface for the M137A2 telescope on the M187A1.

4.11 Workmanship. Workmanship of the M187A1 mount shall be verified in accordance with MIL-F-13926.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Mount, Telescope and Quadrant: M187A1 supports a panoramic telescope and an elevation quadrant for laying the artillery weapon for indirect fire in azimuth and elevation, respectively.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Selection of an applicable level of preservation, packaging, and packing in accordance with MIL-STD-2073-1, Department of Defense Standard Practice for Military Packaging.

c. Packaging Data Sheet SPI 12984689 as applicable (See 6.5).

d. Applicable stock number.

e. Provisions for first article testing.

6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection should be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: RDAR-QEW-A, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1423 in the contract. When the contractor submits inspection equipment designs to the Government for approval, he should give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc).

c. The designs remaining to be submitted and the expected date of submittal.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research Development and Engineering Center (ARDEC) Drawings may also include drawings prepared by, and identified as Edgewood Arsenal,

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Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Packaging data sheet drawings. The following packaging data sheet drawings should be obtained from U.S. Army ARDEC, ATTN: RDAR-EIL-P B455, Picatinny Arsenal, NJ 07806-5000.

SPI 12984689 - Special Packaging Instruction for Mount, Telescope, and Quadrant: M187A1

6.6 Thermal shock. Caution should be exercised during environmental testing to avoid subjecting the mount to thermal shock. A rate of change of 5°F per minute has been shown not to thermally shock the units; however a different rate of temperature change may be used at the tester's discretion.

6.7 Shock verification. The transverse shock requirement of -60 g's and +60 g's was derived from actual test data recorded during firing of the M119 series howitzer at a quadrant elevation of 1244 mils using a zone 8 charge (referred to as "high angle, high charge"). These conditions result in the maximum shock to the howitzer and fire control. Repeated exposure to transverse shock loads at the 60 g level has been shown to prematurely wear the bearings and bearing surfaces of the cant correction worm shaft. In an effort to reduce wear on new units, transverse shock loads of -20 g's and +20 g's are to be used for conformance verification, as this is representative of the majority of the shock loads that the units will see in the field. The full 60 g transverse shock requirement must be used for the First Article Testing units to verify that they are capable of meeting the full requirements.

6.8 Subject term (key word) listing.

Howitzer
M119
Towed Artillery
Fire Control

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR
(1240-2013-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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